SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO5: *Introduction – NumPy Arrays – NumPy Basics- Math – Random – Indexing – Filtering – Statistics – Aggregation – saving data – Data analysis with Pandas – DataFrame – Combining – Indexing – File I/O – Grouping – Filtering – Sorting – Plotting (BL 3)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 **Total Marks:20**

Annexure A

MOCK INTERVIEW QUESTIONS

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

1. You are working as a data analyst in a hospital where patient data is continuously collected and stored in CSV files. However, the dataset contains missing values, inconsistent units, and duplicate entries. The system must process this data in near real-time for reporting. Explain how you would design a solution using NumPy and Pandas to clean, filter, and standardize the data efficiently. Discuss indexing, aggregation, and saving the processed data, along with strategies to handle unexpected data anomalies.

The hospital receives patient records continuously in CSV files. **Pandas** can be used to read, clean, transform, and organize the data, while **NumPy** can perform efficient numerical operations.

First, the CSV file is loaded using pd.read_csv(). The dataset is inspected using head(), info(), describe(), and isnull().sum(). Missing numerical values can be replaced using the mean or median, while categorical missing values can be replaced with the mode. Duplicate records can be identified using duplicated() and removed using drop_duplicates().

Inconsistent units must be standardized before applying machine learning algorithms. For example, temperature values can be converted into a common unit and weight values can be converted into kilograms. Numerical columns can then be normalized or standardized using NumPy or ML preprocessing techniques.

Indexing and filtering are used to select required patient records. For example, patients with abnormal blood pressure can be filtered using Boolean conditions. Pandas groupby() can calculate statistics such as average age, average glucose level, or number of patients in each department.

Unexpected anomalies such as negative age, impossible blood pressure values, or incorrect data types should be detected using range checks and filtering. Invalid records can either be corrected, removed, or flagged for manual verification.

After cleaning, the processed DataFrame can be saved using to_csv(). The cleaned dataset can then be supplied to a machine learning model for tasks such as disease prediction or patient risk classification.

ML relevance: Data cleaning, normalization, outlier detection, categorical standardization, and missing-value handling improve the quality and accuracy of machine learning models.

2. **A financial company provides you with millions of transaction records that must be analyzed to detect unusual spending patterns. Due to system limitations, memory and computation time are critical constraints. Describe how you would use NumPy arrays and Pandas DataFrames to perform filtering, grouping, and statistical analysis efficiently. Explain how you would optimize indexing and aggregation operations, and how you would debug performance bottlenecks or incorrect outputs.**

For millions of financial transactions, memory and processing time are important constraints. NumPy arrays are useful for fast numerical calculations, while Pandas DataFrames provide efficient data manipulation and analysis.

The transaction data can be loaded using Pandas. Only required columns should be selected instead of loading unnecessary data. Appropriate data types such as float32, int32, and categorical types can reduce memory consumption.

Boolean filtering can identify transactions above a particular amount or transactions occurring in unusual locations. Pandas groupby() can be used to calculate total spending, average transaction value, and transaction count for each customer.

Statistical measures such as mean, median, standard deviation, minimum, and maximum can be calculated using Pandas and NumPy. Unusual transactions can be detected using methods such as Z-score or Interquartile Range (IQR). For example, transactions with values significantly higher than the normal spending pattern can be marked as anomalies.

Indexes should be created only when they improve repeated lookup operations. Unnecessary indexes can increase memory usage. Aggregation should be performed using vectorized Pandas and NumPy operations instead of Python loops because vectorized operations are faster.

For debugging performance, execution time and memory usage can be monitored. Operations involving unnecessary loops, repeated DataFrame copying, excessive columns, or inefficient joins should be identified and optimized. Results can also be validated by checking row counts, group totals, and sample records.

ML relevance:

The detected transaction anomalies can become labels or features for machine learning models used in fraud detection and unusual spending prediction.

3. **You are given multiple datasets from different departments (sales, marketing, and customer support), each stored in different file formats. Your task is to combine these datasets into a unified DataFrame while ensuring no loss of critical information and maintaining consistent indexing. Explain your approach using Pandas for file I/O, combining, grouping, and sorting. How would you validate the merged dataset and resolve conflicts such as missing or duplicate keys?**

Datasets from sales, marketing, and customer support may have different formats such as CSV, Excel, and JSON. Pandas provides functions such as read_csv(), read_excel(), and read_json() for importing these files.
After loading the datasets, column names and data types should be standardized. For example, Customer_ID, customer_id, and CustID should be converted into a common column name such as customer_id.
If datasets contain related records, Pandas merge() or join() can be used. For datasets containing similar rows, concat() can be used to combine them. A common customer ID can be used as the key.
Duplicate keys should be checked using duplicated(). Missing keys can be identified using isnull(). Depending on the business requirement, missing keys can be corrected, removed, or retained separately for investigation.
After merging, the dataset should be validated by checking the number of rows, number of unique customers, missing values, duplicate records, and data types. Sorting can be performed using sort_values(), and grouping can be performed using groupby().
For machine learning, the final unified DataFrame can contain useful features such as customer purchase frequency, marketing response, support requests, total spending, and customer activity.
ML relevance:
Combining multiple sources creates a richer feature set and allows machine learning models to learn from sales, marketing, and customer-support behavior together.

4. **A company requires an automated system that reads daily data files, processes them, and generates summary reports along with visualizations. The pipeline must filter relevant data, compute statistics, and produce sorted outputs for dashboards. Describe how you would design this workflow using NumPy and Pandas, including file handling, filtering, aggregation, and plotting. Also explain how you would handle errors such as corrupted files, inconsistent formats, or missing columns.**

An automated data pipeline can be created using **Pandas and NumPy**. Every day, the system reads the newly generated data files from a specified folder.
The pipeline first checks whether the file exists and whether its format is valid. Pandas functions such as read_csv() or read_excel() are used to load the data. The data is then checked for missing columns, incorrect data types, duplicate records, and missing values.
Relevant records are selected using Boolean filtering. Numerical statistics such as mean, median, minimum, maximum, and standard deviation are calculated using NumPy and Pandas. groupby() can be used to generate department-wise, product-wise, or customer-wise summaries.
The processed results can be sorted using sort_values() and saved using to_csv() or to_excel(). Matplotlib can then be used to generate graphs such as bar charts, line graphs, and histograms for dashboards.
Errors should be handled using exception handling. If a file is corrupted, the system should log the error and continue processing other valid files. If required columns are missing, the file should be flagged rather than processed incorrectly. Different date formats and numerical formats should be converted into a standard format.
ML relevance: The same pipeline can generate clean datasets and features automatically for daily or real-time machine learning prediction.

5. You are working on an e-commerce platform where customer behavior data is collected in real time. The dataset contains clickstream data with missing values and inconsistent formats. Explain how you would preprocess and structure this data using Pandas. Discuss filtering, feature extraction, indexing strategies, and how you would prepare the data for downstream machine learning tasks.

In an e-commerce platform, customer behavior is collected as **clickstream data**, including customer ID, product ID, clicks, page views, purchases, price, and timestamp. Since the data may contain missing values and inconsistent formats, **Pandas and NumPy** can be used to preprocess it before applying machine learning.

1. Data Loading

The data can be loaded using pd.read_csv() and inspected using head(), info(), and describe().

2. Handling Missing Values

Missing values can be identified using isnull(). Numerical values can be filled using the median, while categorical values can be replaced with "Unknown".

3. Removing Duplicates

Duplicate clickstream records can be removed using drop_duplicates() to avoid incorrect customer activity counts.

4. Standardizing Data

Inconsistent formats can be standardized. For example, timestamps can be converted using pd.to_datetime(), and text values such as Purchase and purchase can be converted to lowercase.

5. Filtering

Invalid records such as negative prices or missing customer IDs can be removed using Boolean filtering.

6. Feature Extraction

Useful features can be extracted, such as **number of clicks, product views, purchases, total spending, session duration, hour, and day of the week**. These features are useful for machine learning.

7. Indexing and Aggregation

customer_id can be used as an index for efficient customer-level access. The groupby() function can calculate total purchases, total spending, and number of clicks for each customer.

8. Preparing for Machine Learning

Categorical data can be encoded and numerical data can be scaled. Finally, the dataset can be divided into **training and testing sets** and given to a machine learning algorithm.

Conclusion: Pandas and NumPy help clean, filter, structure, and transform clickstream data into useful features for ML applications such as **purchase prediction, customer segmentation, recommendation systems, and churn prediction**.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**25**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand